

Benjamin Garcia

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EDUCATION

University of California, Los Angeles

Bachelor of Science, Computer Science

Los Angeles, CA

Expected Graduation: Jun. 2027

TECHNICAL SKILLS

Languages: TypeScript, Python, Java, C, C++, SQL, JavaScript, HTML/CSS

Frameworks & Tools: Next.js, React, Tailwind CSS, FastAPI, Prisma, Node.js, Redis, PostgreSQL, Supabase, Firestore, JIRA

Cloud & DevOps: AWS Lambda, Palantir Foundry, Docker, Vercel, Git, CI/CD, Linux/Unix, GitLab, Kubernetes

EXPERIENCE

Interface Undergraduate Researcher

Dec. 2025 – Present

UCLA Human-Computer Interaction Research Lab

Los Angeles, CA

- Developing interactive **human-in-the-loop** interfaces in **React**, **TypeScript**, and **Next.js** for exploring and refining AI-generated hypotheses over large-scale knowledge graphs.
- Collaborating with **PhD researchers** to extend a **GraphRAG-based reasoning framework**, designing UI systems that surface intermediate reasoning steps, uncertainty, and evidence for user-driven sensemaking.
- Implementing **multi-agent LLM workflows** via Python services and real-time channels (**WebSockets**), enabling structured, iterative hypothesis generation and rapid experimental iteration for **CHI/IUI/NAACL** research.

Software Engineer Intern

Mar. 2025 – Sep. 2025

Todd Agriscience

Remote, Part-time

- Delivered AI-driven crop insight dashboards for **5–10** clients by architecting full-stack systems in **Next.js**, **Tailwind CSS**, and **Framer Motion**, integrated with **Palantir Foundry** and **AWS Lambda**.
- Developed dynamic data pipelines visualizing **20+** agricultural data streams (soil, irrigation, weather, yield) to enable real-time farm management decisions.

Research and Development Engineer Intern

Jul. 2025 – Aug. 2025

Bonterra

Remote

- Prototyped an **agentic AI** system for nonprofit event analytics using **tool-calling frameworks** and **Model Context Protocols (MCPs)**, modeling attendance, retention, and fundraising across datasets; benchmarked **5+** agent frameworks.
- Developed and tested a multi-tool orchestration layer enabling **autonomous agent collaboration**, improved retrieval latency by **22%**, and validated tool-selection policies for scaling across nonprofit datasets.

Software Engineer Intern

May 2025 – Jun. 2025

TensorStar

Remote

- Engineered real-time agentic UIs with **Next.js**, **Redux**, and **WebSockets**, supporting **100+** concurrent users with sub-**100ms** latency while collaborating with a **10–15**-engineer team across frontend, ML, and infrastructure.
- Built secure credential-submission workflows integrated with **HashiCorp Vault** and a **Python + Redis** backend, unifying authentication across 50+ enterprise data sources (AWS Glue, Snowflake, MongoDB, Postgres, Airflow, dbt, Spark).

PROJECTS

Git Proof | *Next.js, TypeScript, PostgreSQL, Prisma, Gemini 2.5, GitHub GraphQL API*

Dec. 2025 – Present

- Architected a **full-stack developer portfolio platform** transforming GitHub activity into recruiter-ready insights, processing **100+ repositories per user** via a type-safe pipeline integrating the **GitHub GraphQL API**, **PostgreSQL**, and **Prisma ORM**.
- Designed a **3-factor weighted impact scoring algorithm** (0–50 scale) combining logarithmic popularity scaling, recency decay across **5 time tiers**, and project maturity signals to fairly rank diverse portfolios with user-level weighted aggregation.
- Built an **AI-powered insights and content system** using **Google Gemini 2.5 Flash**, generating context-aware project descriptions, professional bios, and personalized portfolio feedback by analyzing repository metadata, documentation quality, and contribution patterns.
- Implemented **security- and reliability-first infrastructure**, including read-only GitHub OAuth scopes, database-stored tokens, complete account deletion with **OAuth grant revocation**, and a hybrid caching strategy with TTL-based freshness checks to handle API rate limits while maintaining data integrity.

Poly Clubs | *Next.js, TypeScript, Supabase, PostgreSQL, Tailwind CSS*

Dec. 2025 – Present

- Built a full-stack review platform for **493 student organizations** using **Next.js 15** and **Supabase**, supporting anonymous reviews with browser fingerprinting and database-level rate limiting to prevent abuse.
- Designed a serverless moderation and data access layer using **PostgreSQL RPC functions** and **Row Level Security**, reducing client-server round trips by **60%** and achieving **10–100x** faster queries via indexed aggregation.
- Architected a **security-first validation model** with multi-layer constraints (client, server, database), integrated **FingerprintJS** for cross-session enforcement, and implemented dynamic **OpenGraph metadata** across **493** statically generated pages.